

Full Length Article

The value of leaked data in online social networks

Nadav Voloch^{*} , Ron S. Hirschprung 

Faculty of Engineering, Department of Industrial Engineering and Management, Ariel University, Ariel, Israel

ARTICLE INFO

Keywords:

Online Social Networks (OSNs)
Data leakage
Privacy valuation
User behavior
Privacy controls
Third-party access

ABSTRACT

Online Social Networks (OSNs) have become one of the central platforms for communication, content sharing, and digital interaction. However, their widespread use raises important concerns regarding the exposure of personal information and the economic value of users' data. While previous research has examined privacy risks and data leakage mechanisms in OSNs, relatively little work has quantified how users value their leaked personal information. This study investigates how users assign monetary value to different types of personal data in OSN environments by proposing an experimental framework that estimates privacy valuation through observable user decisions. To this end, an interactive simulation was developed in which participants received incentives for disclosing specific personal information under different contextual conditions. The framework considers four factors: the type of personal data requested, the platform where the data would be exposed, the object offered as an incentive, and the timing of the incentive. The empirical study involved 249 participants. The results reveal substantial variation in how users value their personal data. Financial information, such as bank account details, was associated with the highest privacy valuation, while disclosure behavior was influenced by platform context and the value of the offered incentive. High-value or socially sensitive products increased participants' willingness to disclose information, whereas low-value incentives had minimal influence. These findings demonstrate that privacy valuation in OSNs, specifically of leaked data, is highly contextual and influenced by both economic and social factors. The proposed framework contributes a practical method for estimating the perceived cost of personal data exposure, and offers insights that may inform the design of privacy-aware systems, consent mechanisms, and data governance policies.

1. Introduction

The rapid growth of the popularity of Online Social Networks (OSNs) has transformed the way individuals interact, share information, and engage with digital content. People use social networks to maintain relationships, share information, and shape their personal identities outward. Thus, these platforms serve as important venues for self-presentation and social interaction, especially among young people (Boyd and Ellison, 2007). The impact of OSNs is bi-directional, where social media use has been linked to both positive and negative psychological outcomes, depending on the usage patterns and individual differences (Kross, et al., 2013).

According to Rivera (2025), Facebook maintains its position as the leading social media platform with approximately 3.065 billion monthly active users worldwide, X (formerly Twitter) reports around 611 million monthly active users globally, reflecting its continued relevance in the social media landscape, and WhatsApp continues to be the most popular

messaging application globally, with over 2 billion monthly active users.

Despite the benefits provided by OSNs, the widespread sharing of personal information raises significant privacy concerns. Users frequently disclose sensitive data, including personal identifiers, location information, financial details, and social relationships. Once shared, this information may be accessed, inferred, aggregated, or exploited by other users, third-party applications, or platform operators. Consequently, the exposure of personal data in OSNs may lead to privacy risks such as identity theft, profiling, reputational damage, and unintended information disclosure.

These risks have motivated a growing body of research focusing on privacy protection and awareness in social network environments. For example, Cirillo, et al. (2023) propose a visual privacy tool designed to help users better understand and manage the visibility of their personal information in social networks. Similarly, Cerruto, et al. (2022) analyze how social network data analysis techniques can reveal potential privacy threats associated with information sharing. These works highlight

^{*} Corresponding author.

E-mail address: nadavv@ariel.ac.il (N. Voloch).

<https://doi.org/10.1016/j.cose.2026.104937>

Received 13 January 2026; Received in revised form 17 April 2026; Accepted 21 April 2026

Available online 22 April 2026

0167-4048/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

the increasing need for mechanisms that help users understand and manage privacy risks when interacting with social platforms.

The inherent tension of conflicts between privacy controls and core OSN features is further amplified by the economic and discursive dynamics of social media platforms. While OSNs provide privacy settings and user control mechanisms, the underlying platform architecture and business models often incentivize data sharing and increased visibility. As a result, privacy controls may coexist with design practices and platform narratives that implicitly encourage disclosure. McDonald and Forte (2021) show that social media companies actively shape public discourse around privacy through platform blogs, policies, and communication strategies, constructing narratives that influence how users interpret privacy expectations and norms. Their analysis suggests that privacy norms in social networks are not purely user-driven, but are partially constructed by platform actors in ways that often align with data-driven business models. Consequently, the tension between privacy protection and platform functionality is not merely technical but also socio-economic, reflecting the competing interests between user privacy and data exploitation.

Naturally, a variety of solutions to address the privacy issue in OSNs have been introduced in the literature. However, while convincing privacy-enhancing technologies (PETs) and policies have been proposed, their adoption remains limited due to usability challenges and varying levels of privacy literacy among users. Dibia et al. (2024) point out that differential privacy is hard to implement and explain, which makes it less widely used. Their research emphasizes the need for user-centered design and clear communication to make these tools more accessible to diverse user groups, including developers and non-technical stakeholders. Prince et al. (2024) explore how privacy literacy significantly enhances users' understanding of companies' data practices and their ability to make informed decisions. Their study suggests that without adequate privacy literacy, users may struggle to navigate and utilize privacy settings effectively, leading to limited adoption of privacy-enhancing measures.

One of the most pressing challenges in this domain is third-party data leakage, where user information, often shared within a closed circle, is inadvertently or intentionally disclosed to external entities. This phenomenon raises critical questions about users' awareness of privacy violation resulting from their own publications, and their willingness to trade privacy for digital benefits or incentives. Zhou and Piramuthu (2015) introduce a conceptual framework for understanding how third-party information leakage occurs, particularly in contexts where data is shared within a closed group but later is disclosed to outsiders either intentionally or unintentionally. It highlights the critical role of user behavior and third-party access in the propagation of sensitive data. Wong et al. (2019) examined the psychological mechanisms behind employees' potential exposure of personal data behaviors and suggests that fostering an organizational ethical climate and a robust information security culture can mitigate such risks. The authors propose a framework to address information leakage issues effectively. Kim et al. (2020) analyzed insider data leaks, emphasizing the predictive behaviors leading to security incidents. The study evaluates risk based on employee actions and proposes strategies to anticipate and prevent data leakage, which is actually an unintended disclosure caused by human factors. While previous research has primarily focused on technical privacy risks or privacy attitudes, the economic valuation users assign to their personal data in practical disclosure scenarios remains insufficiently explored.

Prior research has extensively explored the mechanisms behind privacy leakage in OSNs, highlighting the role of third-party applications, platform vulnerabilities, and user behavior in exposing sensitive personal data (Li, et al., 2015). Studies such as Mitchell and El-Gayar (2022), indicate that OSN users often underestimate the extent to which their information is accessible and exploitable, even when privacy controls are in place. Therefore, the awareness issue is significant, directly deriving the potential for privacy loss.

According to Nayak and Ojha (2019), data leakage is defined as the disclosure of confidential data to an unauthorized person; whether internal or external to the organization. This might occur inadvertently or deliberately. The original definition is by nature an organizational type and was adapted to Social Networks. In Social Networks the definition might be obscure, mainly because it is hard to determine whether the data was meant to be shared or not. The term 'platform' is defined as a means of data sharing, e.g., general platforms like Facebook and Instagram, and specific platforms like ResearchGate. The data that is considered is sensitive from the individual point of view (usually in a personal context), rather than in the context of employees in the organizational case. The involvement of sensitive data makes this a very subjective issue, which is dependent on the individual's preferences and characteristics. A good description of the data leakage problem is presented by Li et al. (2015): Data (or privacy) leakage is defined as the unintended disclosure of private user information in OSNs, even when privacy settings are correctly configured. This occurs due to inherent conflicts between privacy controls and core OSN features such as recommendations, tagging, and mutual connections. The authors highlight that while OSNs like Facebook, Google+, and Twitter provide privacy control mechanisms, they are often insufficient to prevent unintended information disclosure. The core issue lies in the **conflict between privacy controls and the inherent functionalities of OSNs**. In the methodology section of this paper, a leakage criterion is defined, i.e., when data is considered leaked, and not just willingly published. This is done in order to prevent user subjectivity or indifference referring to his personal data being published online. The paper investigates the effectiveness of privacy control mechanisms against privacy leakage from the perspective of information flow. It reveals that the existing privacy control mechanisms do not protect the flow of personal information effectively. The research reviews several OSNs, including Facebook and Twitter, to investigate the problem of Privacy Leakage under Privacy Control (PLPC), which is the phenomenon of private information leakage even when privacy rules are properly configured and enforced. It uncovers the underlying reasons that make privacy control vulnerable from the perspective of information flow. The findings of the research reveal that a series of attacks can take place to access undisclosed personal information. The paper analyzes how private user information can be leaked in OSNs despite correct privacy settings, by identifying attacks such as inferring hidden attributes from friends' data, exploiting cross-site privacy mismatches, and uncovering relationships via recommendations. It models information flows within OSNs and demonstrates how core functionalities like tagging and friend suggestions can unintentionally expose personal data. An analysis was done to conclude necessary conditions and provided suggestions for users to mitigate privacy leakage in OSNs.

In the context of this paper, the term 'data leakage' is used in a broad sense to describe situations in which personal information becomes accessible beyond the user's intended audience within OSNs. This includes not only classical security incidents or organizational data leaks, but also unintended disclosure through platform mechanisms, inference from publicly available information, sharing by third parties, or exploitation of user-generated data by platforms and external entities. While the term "leakage" may traditionally imply a failure in protecting confidential data, in OSN environments the boundaries between voluntary disclosure, unintended exposure, and platform exploitation are often blurred. Therefore, the term data leakage is used here to capture the broader phenomenon of personal information becoming available to unintended recipients.

The overall harm associated with data leakage can be understood as the combination of two main factors: first, the likelihood that a leakage event will occur, and second, the value or sensitivity of the information that is disclosed. The probability of leakage may depend on multiple factors, including users' privacy literacy, disclosure behaviour, platform design, privacy settings, and third-party manipulations. At the same time, even when the probability of leakage is relatively low, the

potential harm may still be substantial if the leaked information is highly valuable or sensitive. The present study focuses on this second component, namely the value of leaked information from the user's perspective. This focus should be viewed as a cornerstone within a broader assessment of privacy risk, since a comprehensive evaluation of data-leakage harm requires consideration of both the probability of exposure and the value of the exposed data.

Evaluating the value of the disclosed data, or, in other words, the cost of data disclosure to the individual, is discussed in the literature. For example, Skatova et al. (2023) investigated individuals' willingness to pay (WTP) to protect various types of personal data across different contexts. Their research employed five evaluation techniques to assess whether people place a positive monetary value on keeping their data private. The results indicate that individuals do assign measurable value to their privacy and make trade-offs based on the type of data and context, which can include political or social considerations. Another research that supports these findings was conducted by D'Annunzio and Menichelli (2022). They analyzed the trade-offs consumers face between monetary benefits and personal data disclosure. Using survey data from Norway, they examined respondents' willingness to share data in exchange for discounts and their willingness to pay to keep data private. The research findings highlight that individuals make nuanced decisions about their data privacy, balancing economic incentives against personal and societal considerations.

Sood et al. (2025) propose a novel solution to a critical privacy issue in Augmented Reality / Virtual Reality (AR/VR) environments. The authors highlight how the use of immersive technologies such as EEG, ECG, or facial dynamic – can result in the unintentional (or malicious) leak of sensitive biometric data. To address this, the authors of the paper introduce a two-stage machine learning (ML)-based framework that classifies and filters biometric signals collected by immersive devices. By identifying and blocking sensitive data before it is transmitted or stored, the framework empowers users with better privacy protection. The method was tested using six real-world datasets and was proven to be effective in protecting data, while achieving high classification accuracy even under signal distortions. Their work is among the first to tackle biometric data leak prevention (not just authentication) in immersive technologies.

Despite extensive research on privacy risks, leakage mechanisms, and privacy protection in OSNs, one important question remains insufficiently addressed: how much do users believe their potentially leaked personal data is worth? While prior studies have examined technical vulnerabilities, privacy controls, and privacy attitudes, fewer studies have provided an empirical behavioral framework for estimating the monetary value users assign to personal data disclosure in realistic OSN-related contexts. To address this gap, this study proposes an experimental framework to estimate privacy valuation from observable disclosure decisions. We implemented an interactive e-commerce simulation in which participants received discount offers in exchange for disclosing different types of personal information under controlled contextual conditions. The framework examines four factors: the type of personal data, the platform of exposure, the product offered as an incentive, and the timing of the offer.

The value of leaked personal information should be viewed as a multidimensional construct rather than a fixed attribute of the data itself. In practice, the perceived cost of disclosure depends not only on the information disclosed but also on the contextual conditions surrounding that disclosure. In OSN environments, these conditions include the type of personal data involved, the platform on which the information may be exposed, the type and value of the incentive offered in return for disclosure, and the timing of the incentive's introduction. Each of these dimensions may influence users' perceptions of sensitivity, risk, and reward, thereby shaping the subjective value they assign to leaked information. For this reason, the present study explicitly incorporates these factors into its experimental framework, allowing a more comprehensive estimation of privacy valuation in realistic disclosure

scenarios.

This study seeks to examine the cost of privacy by assessing the monetary value OSN users attribute to different types of personal data in exchange for rewards. The proposed methodology addresses the illusive nature of this task, e.g., the subjectivity of self-reported privacy costs. Specifically, an experiment was designed to quantify users' perceptions of data sensitivity and their willingness to share personal information under varying probabilities of leakage. By analyzing user responses to different data disclosure scenarios, the study derived insights into the trade-offs between privacy protection and economic incentives. Our findings contribute to a deeper understanding of user behavior in OSNs and support the development of more effective privacy-preserving mechanisms and policies.

The main contributions of this research are as follows:

- A behavioral methodological framework that estimates the monetary value users assign to personal information in OSNs.
- A design and implementation of a realistic interactive simulation to evaluate the data value based on controlled incentives for personal data disclosure.
- Empirical analysis of how privacy valuation varies according to four contextual factors: data type, platform of exposure, product offered, and timing of the incentive.
- Quantitative empirical evidence showing that users' privacy valuations are highly contextual and influenced not only by data sensitivity but also by platform environment and incentive structure.

The remainder of this paper is organized as follows. Section 2 reviews related work on privacy risks, data leakage, and privacy valuation in OSNs. Section 3 presents the proposed methodology and experimental framework for estimating the value of personal data. Section 4 describes the empirical study, including the experimental design, participants population, and data collection process. Section 6 discusses the implications of the findings and highlights limitations and future research directions.

2. Background and related work

While prior studies have focused on privacy controls, technical leakage mechanisms, or tools that help users manage privacy settings, an important question remains underexplored: how much do users themselves value their leaked personal data when confronted with disclosure decisions? In other words, when users are offered incentives or benefits in exchange for sharing personal information, what implicit value do they assign to their data? Understanding this valuation is critical for designing privacy-aware systems, improving consent mechanisms, and informing regulatory frameworks that aim to protect user data in digital ecosystems.

The existing literature on privacy risks and data exposure in Online Social Networks (OSNs) can be broadly categorized into several research streams. First, a group of studies focuses on technical mechanisms of privacy leakage, including inference attacks, third-party applications, and architectural vulnerabilities in OSNs. A second stream investigates privacy control mechanisms and privacy-enhancing technologies, proposing models that aim to mitigate unintended disclosure or improve user awareness. A third category examines behavioral and economic aspects of privacy, analyzing how users perceive, evaluate, and trade personal information under different contextual conditions. Finally, recent research explores data valuation and data pricing frameworks, investigating how personal data can be economically valued or traded within digital ecosystems. Organizing the literature into these categories makes it easier to identify both the contributions and the limitations of existing work, as well as the gap addressed by the present study: the lack of empirical frameworks that directly measure the economic value users assign to their personal leaked data in OSN environments under realistic disclosure scenarios.

Krishnamurthy and Wills (2009) is a relatively early paper that generally handles the leakage of personal information from Social Networks. It refers to the architectural risks of using social network platforms with sensitive personal information and introduces data leakage issues. The main vulnerabilities of using third-party HTTP servers are shown, and specifically leakage from third-party applications. The paper also shows prevention solutions that are mainly blocking such applications, or architectural blocking of certain information packets that are known to originate in these applications.

Tomy and Pardede (2016) investigated OSNs' third-party applications with the permission-based access control (PBAC) model, with focus on Facebook. They showed that the PBAC model can enhance user awareness regarding privacy risks of data passed on to third party applications and allow users who are more concerned about their privacy the ability to control the release of such information to such applications. The paper shows an experimental evaluation that is compared to other known models in terms of protection and awareness, such as Privacy by redesigning the UI model, user notification model, and other models. Mitchell and El-Gayar (2022) surveyed privacy in social networks in respect to many of its aspects, including third-party application data leakage in OSN. They also referred to the background of the problem and its generality of disclosing sensitive information, and cover the possible risks such as identity theft and controlling a user's personal information. Finally, in their paper they survey several methods of privacy protection, such as encryption and authentication. Molok et al. (2018) dealt with data leakage from OSNs, in reference to the organizational aspect. They focused on all sorts of data leakage issues in OSN, including third-party applications and links to external sites that are presented as one of OSN's functions and potential risks. The main third-party applications investigated consist of online games, and an organizational solution in which employees would be able to access Facebook at work, but access to external links, games, and applications would be restricted. Another organizational solution that was presented was raising employee awareness to this problem with specific instructions and courses on this issue. The problematic "sensitive" organizational data comprises trade secrets, intellectual property or personal details of employees. The exposure of these may result in a loss of competitive advantage and reputation, loss of client trust, and additional damage. The experimental evaluation included both employees and management across four organizations, leading to the development of a self-assessment maturity framework that evaluates an organization's capability to mitigate sensitive information leakage through Online Social Networking (OSN). The tool assesses seven key factors, ranging from policy integration to employee awareness in order to determine the organization's maturity level, from 'Reactive' to 'Integrated', in managing OSN-related leakage risks. The framework also provides development of an OSN security strategy as a tool to guide and improve the mitigation of OSN security issues in organizations.

Despite extensive research on privacy risks, information leakage, and privacy controls in OSNs, relatively little work has quantitatively examined how users themselves economically value their personal information when confronted with disclosure incentives. In particular, while many studies analyze technical leakage mechanisms, privacy settings, or behavioral attitudes toward privacy, fewer examine the monetary trade-offs users face when deciding whether to disclose personal data in real-world contexts. This study addresses this gap by proposing a framework for estimating the monetary value that users implicitly assign to their leaked personal data in OSN environments. Using an interactive e-commerce simulation in which participants receive discount offers in exchange for sharing different types of personal information, we measure how disclosure decisions vary across data types, platforms, incentives, and contextual conditions. The results provide empirical insights into the economic valuation of personal data and highlight the contextual nature of privacy decision-making in OSNs. Praveena and Smys (2016) surveyed the issues of data privacy in OSNs in general, including data-leakage, and specifically the problematic

aspects of third parties who use social network users' private data. The main solutions that the paper surveys are anonymization solutions, while emphasis is placed on the k -anonymity solution for the use and release of personal OSNs' data, that can be transferred outside the network. Nayak and Ojha (2019) surveyed the issue of data leakage detection and prevention. In their paper they defined data leakage in an organizational context, introduce a prevention system characterized by data state, deployment location, and detection approach, review the phases of data leakage detection and prevention (DLDP), and outline key challenges and future research directions.

Lindamood et al. (2009) explored privacy leakage in OSN's, that is created by inferring private information from non-private data. This paves the way to different attacks, and the paper focuses on how to launch inference attacks using released social networking data to predict undisclosed private information about individuals. The paper also explores the effectiveness of possible sanitization techniques that can be used to combat such inference attacks under different scenarios. The results of the research indicate that removing different details and friendship links is the best way to reduce the threat. This approach is limited because of the deliberate nature of exposure the users seek in these platforms. Srivastava and Geethakumari (2013) introduced a model to ensure privacy in unstructured data, which is also designed for OSNs. The model is called Privacy Armor, and while it is not yet implemented in their research, it is suggested as a theoretical and practical model to deal with the problem of privacy leakage in OSN. In their research they describe and calculates a privacy metric to measure the privacy of the user's profile using the naive approach. This is done by an analysis of a survey that was conducted in the empirical study as an evaluation of how well people understand privacy in OSN. The results were used to shape the model that is aimed to solve the users' privacy leakage problem of users in OSN.

Hirschprung et al. (2016) describe a method to evaluate the value of privacy. This is done with an algorithmic evaluation methodology named VOPE (Value of Privacy Estimation), that is experimentally validated in the research. The paper includes an experimental evaluation, consisting of a user study in which participants were asked to accept or reject hypothetical proposals offering a discount on a purchase in exchange for consent to disclose some transactional data to an e-commerce site. Voloch et al. (2021) proposed a novel approach to enhancing privacy in OSNs by integrating access control and information flow control in trust-based mechanisms. The authors identified the limitations of existing privacy models, such as access control systems and role-based security models, which do not sufficiently protect users from unintended data exposure. Their model classifies users' connections into roles, assesses trustworthiness based on public attributes (e.g., number of friends, account age, and friendship duration), and utilizes information flow analysis to determine potential risks of data leakage. Through a combination of trust evaluation, role-based access permissions, and graph-based information flow control, the model dynamically regulates information sharing to ensure only trusted users are given access. Experimental evaluations using real and synthetic OSN data validate the model's effectiveness, demonstrating a strong correlation between algorithmic decisions and user privacy preferences. Their work provides an advanced framework for safeguarding privacy while maintaining meaningful social interactions in OSNs.

Acquisti et al. (2013) investigated how individuals assign monetary value to their personal data. Through field experiments based on behavioral economics, they discovered that people's valuations of privacy are highly context dependent. The study revealed that individuals often demand significantly more money to relinquish their privacy than they are willing to pay to protect it, highlighting inconsistencies in privacy valuations. These findings suggest that privacy preferences are malleable and systematically shaped by choice architecture, including the framing of disclosure options, the timing of incentives, and contextual cues embedded in the decision environment - raising questions about the effectiveness of market-based approaches to privacy. Despite

its thorough taxonomy of leakage mechanisms, the paper does not empirically assess how such vulnerabilities translate into perceived costs or risks for users. In particular, it does not quantify how users value the potential consequences of data exposure, nor does it provide behavioral or economic measurements that connect technical leakage scenarios to users' disclosure decisions. As a result, a gap remains between the identification of theoretical vulnerabilities and their practical impact on users' privacy-related behavior. Moreover, an important concept emphasized in Acquisti et al.'s paper is the information asymmetry between users and platforms. In digital environments, platforms typically possess significantly more information about the potential uses, aggregation possibilities, and long-term implications of personal data than individual users. As a result, users often make disclosure decisions under incomplete information regarding the risks associated with their data. This asymmetry can influence how individuals value their personal information, potentially leading to underestimation or inconsistent evaluation of privacy risks. In the context of the present study, participants evaluate disclosure scenarios without full knowledge of how data might be aggregated or exploited in real-world OSN ecosystems. Therefore, the estimated privacy valuations observed in the experiment should be interpreted as perceived short-term valuations rather than complete economic valuations, reflecting the information constraints typical of real-world user decisions.

Sunet al. (2025) propose a data pricing framework for Mobile CrowdSensing (MCS) that addresses two challenges: selecting reliable data from sellers whose quality is initially unknown, and designing fair compensation that takes into account device costs and privacy risks. The authors introduce MAB-RS, a multi-armed bandit algorithm to recruit sellers based on learned data quality, and MAB-RSP, a pricing mechanism modeled as a two-stage Stackelberg game. In MAB-RSP, the platform (acts as a leader) sets the payment per unit of data value, and the sellers (act as followers) decide how much quality to provide. The sellers encounter convex cost functions (reflecting increasing marginal costs of better data), while the platform maximizes its utility by balancing data quality gains against payment expenditures. The analysis shows that this model guarantees a unique Stackelberg Nash equilibrium, ensuring both sides maximize profits under optimal strategies. Experiments with real taxi trajectory data demonstrate that the pricing method effectively recruits high-quality sellers, improves revenue, and balances the cost/-profit tradeoff. The platform avoids overspending while sellers are compensated fairly for higher quality contributions.

Galperti and Perego (2023) modeled the economic implications of privacy protection on data valuation. The authors explored scenarios where consumers can choose whether to disclose their data to intermediaries that influence market prices. The study underscores the complex interplay between privacy rights and economic outcomes, suggesting that privacy regulations can have unintended consequences on market behavior and consumer welfare. The privacy evaluation in this paper is limited in scope; it primarily focuses on theoretical constructions and linear programming duality to value data under different privacy regimes. It also lacks empirical validation or behavioral analysis, whereby important questions about how real-world users might actually behave under verifiable or unverifiable disclosure mechanisms, and about how platforms might adapt in practice beyond stylized assumptions were left open. This lack of validation creates a gap between theoretical valuation and practical, measurable privacy impact.

Acquisti et al. (2016) explain that privacy is valued both intrinsically - as a fundamental individual right, and instrumentally - as a means to achieve other objectives such as avoiding discrimination, maintaining social boundaries, or protecting economic interests. They also cover market dynamics where firms benefit from consumer data, often without direct compensation to users, and examine models where privacy regulation can shift power and welfare between data holders and subjects. The authors conclude by emphasizing the need for a nuanced, context-aware approach to privacy economics that recognizes both market and non-market (ethical, legal, psychological) dimensions.

Although the paper spans a wide range of theoretical perspectives from economics, psychology, and policy, it does not synthesize these into a cohesive evaluative methodology, limiting its utility for empirical or policy-oriented privacy assessments. This leaves a gap in translating theoretical insights into actionable metrics for evaluating privacy protections in concrete digital environments.

John et al. (2011) investigate how environmental cues influence individuals' decisions to disclose personal information. They demonstrated, via four experiments, that people are more likely to share sensitive data when prompted by contextual factors, such as the setting or framing of questions that are unrelated or even inversely related to the actual risks of disclosure. The study finds that activating privacy concerns beforehand can mitigate this effect, highlighting the malleability of privacy preferences and the importance of context in privacy-related decisions. A limitation of the privacy evaluation in this paper is its focus on experimental settings with stylized consumer choices, which may not fully capture the complex trade-offs and long-term consequences of privacy decisions in real digital ecosystems. While the study provides valuable behavioral insights, it does not assess how disclosed information can be aggregated, inferred, or exploited over time, leaving a gap in understanding the cumulative and systemic privacy risks users face beyond initial disclosures.

Recent work has also explored the economic and behavioral aspects of personal data valuation in digital ecosystems. For example, the study by (Watson, et al., 2022) investigates how individuals perceive the value of their personal data within digital platforms and highlights the complexity of quantifying privacy trade-offs in real-world contexts. Although their work approaches the problem from a different methodological perspective, it similarly emphasizes the importance of understanding how users evaluate personal data in environments where economic incentives and platform dynamics interact. Our study complements this line of research by providing an experimental framework that directly measures users' willingness to disclose specific types of personal data under controlled incentive structures. To provide a clearer comparison between existing approaches, Table 1 summarizes representative studies according to their research focus, methodology, application context, and key strengths and limitations.

Despite the extensive research done in these above-mentioned papers, several key gaps remain: a) There is still a lack of unified and clear evaluative frameworks. Many theoretical papers span multiple domains but do not synthesize their insights into operational methodologies for assessing real-world privacy risks; b) There is limited empirical validation. Several models and frameworks lack user studies, real-world testing, or behavioral analysis that could confirm their effectiveness outside of simulations or theoretical assumptions; c) There is insufficient quantification of privacy costs. Namely, there is a scarcity of robust, user-focused evaluations that measure how much data leaks actually impact individuals or how users perceive the cost of privacy loss; d) There are gaps in cumulative and systemic risk understanding. The research often emphasizes one-time decisions or isolated leaks, neglecting the compounding nature of data disclosures over time; e) Usability barriers when adopting PETs. While privacy-enhancing technologies are available, they are often too complex or poorly communicated to accommodate broad user adoption, particularly among non-technical users; and f) There is a lack of market and incentive-aligned solutions: Very few studies connect privacy behaviors to concrete economic incentives or explore how platforms might exploit or mitigate privacy dynamics in practice. Inspired by the above-mentioned works, the proposed approach is designed to evaluate the value of sensitive data that is leaked from OSNs. While many prior studies have focused on technical leakage mechanisms, privacy controls, or theoretical models of privacy valuation, only a few works empirically estimate how users economically value their personal data when confronted with concrete disclosure incentives. The present study addresses this gap by introducing an experimental framework that measures users' willingness to trade personal data for tangible benefits under controlled conditions. By

Table 1
Comparison of related work on privacy and data leakage in OSNs.

Study	Research Focus	Methodology	Application Context	Strengths	Limitations
(Krishnamurthy and Wills, 2009)	Third-party information leakage in OSNs	Network traffic analysis	Social networking platforms	Early identification of architectural leakage risks	Focuses mainly on technical infrastructure, not user behavior
(Lindamood, et al., 2009)	Inference attacks on social network data	Data mining and inference modeling	Public OSN datasets	Demonstrates how hidden attributes can be inferred	Does not evaluate users' perception of risk
(Srivastava and Geethakumari, 2013)	Privacy protection model (Privacy Armor)	Privacy metric and conceptual model	OSN user profiles	Introduces measurable privacy metric	Lacks empirical validation
(Acquisti, et al., 2013)	Economic valuation of personal data	Behavioral economics experiments	Consumer privacy decisions	Demonstrates context-dependent privacy valuation	Does not examine OSN-specific leakage mechanisms
(Galperti and Perego, 2023)	Economic implications of privacy regulation	Economic modeling	Data markets	Explores relationship between privacy and market behavior	Theoretical model without behavioral validation
(Watson, et al., 2022)	Perceived value of personal data	Empirical user study	Digital platforms	Highlights complexity of privacy valuation	Does not isolate contextual factors affecting disclosure
(Hirschprung, et al., 2016)	Privacy valuation through incentives	Experimental algorithmic framework	E-commerce data disclosure	Provides structured evaluation method	Limited focus on OSN context

combining behavioral experimentation with a simulated disclosure environment, this research binds theoretical models of privacy valuation to observable user decision-making. The following sections present a methodology that provides a framework for this purpose, an empirical study demonstrating its implementation, and the interesting results that were obtained.

3. Methodology

This study proposes a methodology for estimating the cost of leaked information by conceptualizing privacy loss as an economic transaction. The core idea is that when users are asked to disclose personal data under varying conditions, their acceptance or rejection of incentives reflects the implicit monetary value they assign to the potential leakage of that data. Rather than relying on abstract measures of privacy attitudes, the current approach derives quantitative estimates of privacy valuation from users' observable choices.

The methodology is inspired by behavioral economics and decision theory, recognizing that while users may not always act in strictly rational ways, their decisions still exhibit consistent patterns under structured conditions. Therefore, instead of assuming full rationality, the proposed methodology relies on observable choice behavior in a controlled setting to estimate privacy valuation. Specifically, the iterative and indirect design of the simulation circumvents the limitations of irrational or biased decision-making by focusing on disclosed preferences rather than self-reported attitudes. The cost of leakage is operationalized as the minimal compensation users are willing to accept for potential exposure of their data, shaped by contextual factors such as data type, platform, incentive, and timing. This allowed us to capture users' implicit valuation of privacy even in the presence of cognitive biases, framing effects, or bounded rationality.

With this approach we were able to find both lower and upper bounds (if they exist) of privacy valuation through an iterative negotiation process, producing a range that reflects the user's specific cost of data leakage or unintended disclosure. The framework therefore connects the abstract concept of privacy loss with measurable behavioral outcomes, providing a structured foundation for the empirical study described in the section below.

3.1. Preliminaries

Information leakage in OSNs refers to situations in which user information becomes accessible to unintended entities through platform mechanisms, user actions, or third-party access. This may occur through several mechanisms, including unintended disclosure by users, sharing by other users, inference from publicly available information, or exploitation by platforms or external services. In this paper, the event 'leakage' is used as an umbrella concept encompassing both accidental

disclosure and broader forms of unintended exposure of personal information within OSNs.

Information leakage in OSNs can occur in several ways:

- Private information may be deduced from the information that was shared and through auxiliary sources.
- Leaked (intentionally) by one of the authorized information destinations.
- Data breach occurred due to a malicious attack.
- Unintentional (accidental) sharing of the data to unwanted entities (e.g., sharing a post with everyone on Facebook when the intent was to share it with one friend).
- Unintentional (accidental) data sharing by a friend with unwanted entities (e.g., forwarding a message from a private WhatsApp group).
- Image that contains sensitive data (e.g., a photo from a dinner at a restaurant while the credit card appears on the table).
- A video that includes sensitive data (e.g., a street address appearing on a letter viewed in the video).

Obviously, the way in which the information is leaked is not expected to influence the value of the information. However, there are four factors which may affect the value, as detailed below:

- Type of Personal Information Disclosed** – Participants are asked to consider exposing different types of personal data. Example of the categories include basic identifiers (e.g., name), contact details (e.g., phone number, email), and sensitive financial information (e.g., bank account details).
- Platform of Exposure** – The disclosure context varied by the level of publicity associated with different online platforms. For example, the platforms ranged from highly public (open to all users by default) to relatively private (restricted to closed groups).
- Type of Item Rewarded** – To simulate real-world incentives, participants were offered discounts on items in exchange for disclosing personal data. The items varied in price and social sensitivity. For instance, both low-cost items (e.g., stationery) and more socially sensitive products (e.g., personal accessories) were included.
- Timing of the Discount Offer** – The experimental flow was manipulated to test the influence of the timing of the incentive. In one, the negotiation on the discount occurred immediately after each item was presented while in the other, the negotiation took place only after all the items had been displayed.

3.2. General approach

To assess users' valuation of privacy, an interactive e-commerce simulation was developed in which participants were offered discount coupons in exchange for sharing various types of personal information.

The online store emulated a realistic shopping experience, and participants' choices were influenced by contextual variables such as the sensitivity of the requested data, the publicity level of the platform, the product category, and the timing of the discount offer. This setup allowed us to measure the users' privacy preferences dynamically, by using a gamified negotiation process.

To clarify how the proposed methodology operates in practice, consider the following example scenario from the simulation environment. A participant browsing the simulated e-commerce store selects a product, such as a smartphone priced at 500\$. The system offers the participant a discount in exchange for disclosing a specific type of personal information, for example an email address that would be displayed on a given platform (e.g., Facebook). In the initial step, the system proposes a discount offer (e.g., 20%) in exchange for sharing the data item. If the participant accepts the offer, the system records this value as an upper bound for the participant's privacy valuation under that scenario. If the participant rejects the offer, the system increases the discount iteratively (e.g., 30\$, 40\$, 50\$) until the participant accepts the trade-off or the predefined limit is reached. Through this iterative negotiation process, the system estimates the minimum compensation that a participant requires in order to disclose the requested personal data. The accepted value therefore represents the participant's implicit valuation of the potential privacy loss associated with exposing that information under the given contextual conditions (data type, platform, product, and timing). By repeating this process across different combinations of experimental factors, the study constructs a behavioral estimate of how users economically value different types of personal information in Online Social Networks.

In this study, the cost of leaked information was operationalized as a monetary value that users implicitly assign to their personal data when facing disclosure scenarios. Instead of treating privacy as an abstract concept, the proposed methodology adopts an economic perspective: information leakage is modeled as a market-like transaction in which personal data is exchanged for tangible benefits. The underlying assumption is that the willingness to disclose, or withhold, information reveals users' perceived cost of potential leakage. By observing acceptance thresholds and negotiation behavior under different contextual factors, it is possible to estimate both lower and upper bounds of this perceived cost. Given that the process converges, and that the difference between the lower and the upper bounds is not significant (singular

rather than a range), the cost of leakage was revealed. This approach integrates behavioral economics with privacy research, which enabled us to progress beyond purely theoretical definitions of privacy loss and to capture how individual's tradeoff the risk of exposure and the perceived reward in practice.

At the beginning of the process, all the required variables were initialized, including the stage variable, which determines the timing of the discount negotiation: The value of the stage was either *during* (negotiation took place immediately after each item was introduced) or *after* (negotiation began only after all the items were presented). If the stage was set to *after* (meaning *during* is false), the algorithm first established an initial bound, for the prices users are willing to accept in exchange for exposing their data on different platforms, and if not, there was no initial bound. After this initial round, the process continued with the *SI* (Sale Item) values for negotiation, according to the user's first decision. If the stage was set to *during*, the negotiation process occurred immediately at the beginning, upon presentation of each item.

Under the *after* condition, participants first viewed the full set of products without any offers. The system initiated negotiations only during the second round, beginning with the pencil item. Fig. 1 illustrates the starting screen of the negotiation phase under the *after* condition.

With this setup, the participant, browses the e-commerce website and receives discount offers in exchange for sharing personal data items, as defined in the previous section.

3.3. The evaluation process

We designed a controlled experiment to assess the monetary value that OSN users assign to their personal information. The experiment employed a factorial design involving the four main factors that were mentioned above.

Each participant was randomly assigned to one experimental condition: meaning a specific combination of factor values (e.g., type of data, platform, product, and timing), and their responses concerning their willingness to disclose personal information in return for the incentives offered were recorded. The notation for the variables of the process are defined as depicted in Table 2, in order to formalize the process.

Then, a matrix that includes all values of the four experimental

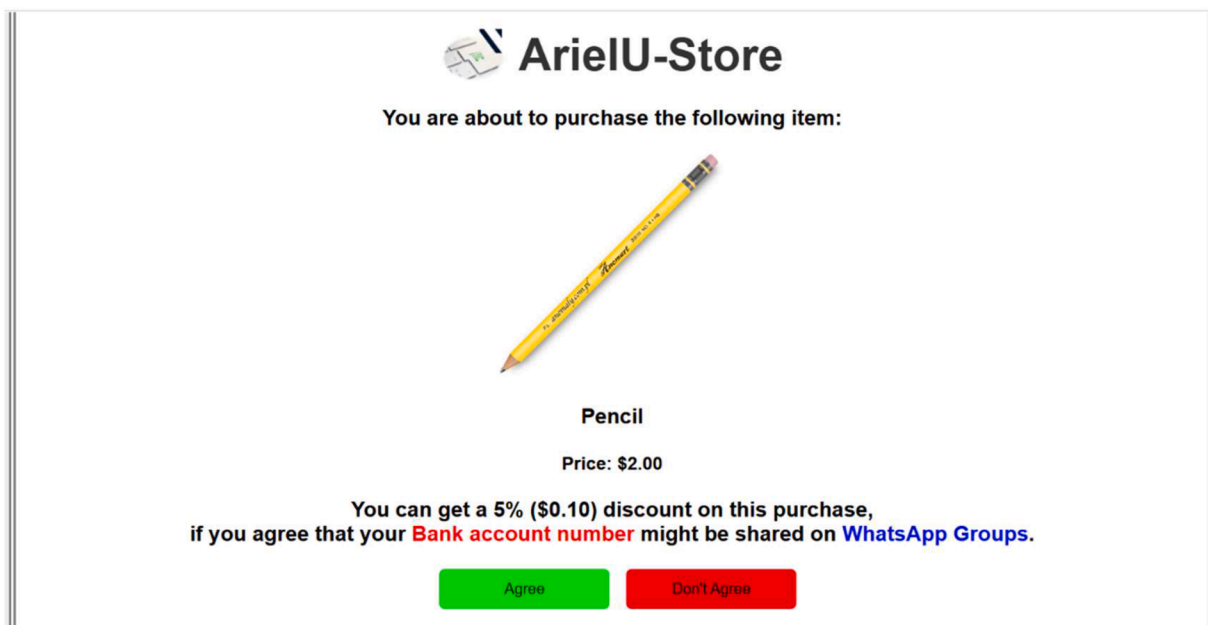


Fig. 1. The initial screen for buying a pencil in the *after* scheme, which was the beginning of the second round.

Table 2

The notations of the variables in the privacy price evaluating simulation and their descriptions.

Notation	Description
PL	Platform: the type of platform (OSN) the exposure is on
SI	Sale Item: the item being sold
DI	Data Item: the type of personal data that will be exposed in the network
v_{ij}	Initial value of the offering for $i = PL, j = DI$
R_{ij}	Random choice for $i = PL, j = DI$
$accepted$	Boolean, the participant accepted/did not accept the offer.
$during$	Boolean, stage of purchase at which the discount is given.
t_{ij}	Number of times the offer will be suggested – to set upper and lower bounds.
Δ_{max}	Maximal value increase offered to participants.
Δ_{min}	Minimal value decrease offered to participants.
$price_{accepted}$	The price the user accepted for the risk of his data being leaked
B_{upper}	Upper bound of the estimated value of privacy
B_{lower}	Lower bound of the estimated value of privacy
R_t	Random int (0–3) for a minimal 0 times and maximal 3 times of increasing/decreasing the offer

factors was defined: the type of personal data, the platform of exposure, the type of product offered, and the timing of the discount negotiation. This matrix is presented in Table 3 in the next section (the empirical study).

The entire process of the e-commerce simulation is described in Algorithm 1 below.

Algorithm 1 formalizes the “negotiation” process through which the process reveals the users’ implicit valuation of their personal data. It represents the core mechanism underlying the interactive e-commerce simulation described in the methodology. The algorithm receives the four experimental factors as input: Platform, Sale Item, Data Item, and Timing, as well as the initial offer values and step sizes for increasing or decreasing the discount. It then iterates through all combinations of factor values, offering discount coupons in exchange for data disclosure and records the users’ responses. The procedure begins by determining whether the negotiation occurs during product presentation or after all items have been displayed. Under the *after* condition, the algorithm first establishes baseline bounds for users’ acceptable offers by randomly sampling from the factorial matrix. Under the *during* condition, negotiation begins immediately with the first item. For each offer, the algorithm checks whether the participant accepts or rejects the proposed trade-off. If the offer is rejected, it is iteratively adjusted, either by decreasing or by increasing the discount value, until an upper or lower bound is reached. These bounds represent, respectively, the maximum and minimum monetary values for which a participant is willing to disclose personal information in a given context. The output of the algorithm consists of two key measures: the accepted price for each factor combination and the corresponding upper and lower privacy valuation bounds. This process enables the estimation of participants’ subjective

data valuation under controlled and repeatable conditions.

Fig. 2. depicts an example of an advanced stage of a ‘negotiation’ for the sale item (SI) which in this case is a smartphone.

The entire code, images and Full Stack project of the simulation are available on GitHub (Voloach, 2025a).

4. The empirical study

In the empirical study, a planned factorial experiment was implemented in accordance with the methodology outlined above. The goal was to quantify how OSN users value the disclosure of personal information when facing concrete incentive structures. The experiment included four factors as described above. Table 3 presents the complete factorial matrix of experimental conditions, combining the different platforms of exposure (Facebook, X/Twitter, and WhatsApp Groups) with the various personal data types (name, email address, bank account details, phone number, and home address). In order to examine how incentive magnitude and context influence disclosure decisions, in addition to the platform and data type, the experiment included various types of products offered, ranging from low-value items (e.g., pencils, chocolates) to high-value or socially sensitive products (e.g., smartphones, lingerie, adult toys). The timing of the discount offers also varied, with some participants receiving incentives immediately after each product presentation (*during* condition) and others only after viewing all products (*after* condition), allowing the study to capture behavioral differences related to decision timing and cognitive anchoring.

4.1. Participants and experiment environment

In total, there were 249 valid participants. Participants were recruited through the crowd-sourcing mechanism Amazon Mechanical Turk (MTurk), and directed to the external cloud-based platform for experience management Qualtrics, and from Qualtrics, after filling in survey details in the simulation website platform. The simulation platform was implemented as a lightweight client-server web application using standard front-end technologies (HTML5, CSS, Bootstrap, and jQuery) to simulate a realistic e-commerce environment. Interactive elements, including product selection, dynamic discount negotiation, and user-decision logging, were handled by custom JavaScript functions (e.g., randomized negotiation loops and data-collection arrays) executed entirely on the client’s side, while PHP scripts processed and stored the submitted responses. Only workers aged 18 or older could participate, and were asked to provide their consent. To prevent duplicate participation, a custom MTurk qualification was assigned to all workers who completed the first batch, and only workers who did not hold this qualification were allowed to access subsequent HITs. Completed responses were screened automatically in the Qualtrics survey using attention-check items, logical-consistency tests, and validation rules included in the questionnaire. Observations were disqualified if participant’s attention checks failed, if they showed inconsistent or illogical responses, or exited the experimental task without reaching the confirmation code. After applying these exclusion criteria, a total of 249 valid responses remained. Of all the participants, 12% were females and 88% were males. With respect to age, 19% were 18–25 years old, 29% were 26–30 years old, 47% were 31–40 years old, 4% were 41–50 years old, <1% were 51–60 years old, and 1% were above 60 years old. Regarding education, 81% had a bachelor’s degree, 12% had a master’s degree, 4% had a high school diploma, 2% had an associate degree, and fewer than 1% reported either no schooling, some schooling without a diploma, or a doctorate degree. Most participants (73%) were employees, 25% were self-employed, and fewer than 1% were homemakers or unemployed.

The experiments were designed to reflect realistic privacy trade-offs commonly encountered in digital platforms, where users may be offered benefits in exchange for personal information. The selected platforms (Facebook, X/Twitter, and WhatsApp Groups) represent distinct

Table 3

All of the different combinations of the values of factors PL , DI .

PL (Platform)	DI (Data Item)
Facebook	Name
Facebook	Email address
Facebook	Bank account details
Facebook	Phone number
Facebook	Home address
X (previously Twitter)	Name
X (previously Twitter)	Email address
X (previously Twitter)	Bank account details
X (previously Twitter)	Phone number
X (previously Twitter)	Home address
WhatsApp Groups	Name
WhatsApp Groups	Email address
WhatsApp Groups	Bank account details
WhatsApp Groups	Phone number
WhatsApp Groups	Home address

Algorithm 1

Privacy value estimation game.

```

input:  $v_{i,j}$ ,  $accepted$ ,  $during$ ,  $t_{i,j}$ ,  $\Delta_{max}$ ,  $\Delta_{min}$ ,  $price_{accepted}$ ,  $B_{upper}$ ,  $B_{lower}$ ,  $R_t$ 
 $B_{lower} \leftarrow 0$ 
 $B_{upper} \leftarrow \infty$ 
 $price_{accepted} \leftarrow 0$ 
 $accepted \leftarrow false$ 
if  $during = false$ 
  for all the values of  $SI$ 
     $R_{i,j} \leftarrow$  Random value from the cells  $c_{i,j}$  from Matrix ( $PL \times DI$ )
    Offer a coupon with the value of  $v_{i,j}$ 
    if  $accepted = true$ 
       $price_{accepted} \leftarrow v_{i,j}$ 
       $B_{upper} \leftarrow price_{accepted}$ 
    else
       $B_{lower} \leftarrow v_{i,j}$ 
       $B_{upper} \leftarrow v_{i,j} + \Delta_{max}$ 
    end
  end
for all the values of  $SI$ 
     $R_{i,j} \leftarrow$  Random value from the cells  $c_{i,j}$  from Matrix ( $PL \times DI$ )
    Offer a coupon with the value of  $v_{i,j}$ 
    if  $accepted = true$ 
       $price_{accepted} \leftarrow v_{i,j}$ 
       $B_{upper} \leftarrow price_{accepted}$ 
       $t_{i,j} \leftarrow R$ 
       $index \leftarrow 0$ 
      while  $index < t_{i,j}$ 
        Offer  $v_{i,j} - \frac{index}{t_{i,j}} \Delta_{min}$ 
        if  $accepted = false$ :
           $B_{lower} \leftarrow price_{accepted}$ 
        else
           $price_{accepted} \leftarrow v_{i,j}$ 
        end
         $index \leftarrow index + 1$ 
      end
    else
       $B_{lower} \leftarrow v_{i,j}$ 
       $B_{upper} \leftarrow v_{i,j} + \Delta_{max}$ 
       $t_{i,j} \leftarrow R$ 
       $index \leftarrow 0$ 
      while  $index < t_{i,j}$ 
        Offer  $v_{i,j} + \frac{index}{t_{i,j}} \Delta_{max}$ 
        if  $accepted = true$ 
           $price_{accepted} \leftarrow v_{i,j} + \frac{index}{t_{i,j}} \Delta_{max}$ 
           $B_{upper} \leftarrow price_{accepted}$ 
        end
         $index \leftarrow index + 1$ 
      end
    end
  end
return  $B_{lower}$ ,  $B_{upper}$ 

```

disclosure contexts, ranging from semi-public to group-oriented. Likewise, the chosen data items and products were selected to capture both variation in perceived sensitivity and variation in incentive value. This design increases the relevance of the experiment to the real-world OSN disclosure decisions while maintaining experimental control.

4.2. Results

The empirical evaluation was designed to examine how users value the disclosure of personal data under controlled yet realistic decision-making conditions. Each participant interacted with the simulated e-commerce environment and was exposed to disclosure scenarios varied with four factors: personal data type, platform of exposure, sale item, and timing of the offer. Participants' responses to discount offers in exchange for data disclosure were recorded and translated into privacy valuation estimates. The evaluation combined a factorial experimental

design with an iterative behavioral elicitation process, enabling a systematic comparison of disclosure decisions across multiple contextual conditions.

To produce participants' privacy valuations comprehensively, in this study, we employed two complementary estimation measures: (1) the percentage of the discount offered to the user, and (2) the absolute monetary value of the discount. The absolute value measure reflects participants' concrete willingness to accept specific amounts of money in exchange for disclosing personal information, thus enabling direct economic interpretation of privacy valuation. However, since the products that were offered varied widely in their base prices, e.g., a smartphone versus a pencil, absolute values alone cannot capture relative behavioral patterns across conditions. Therefore, percentage-based estimates were also calculated to normalize price differences and to assess relative disclosure thresholds independently of the absolute value of the item. This dual approach enables a distinction between users'

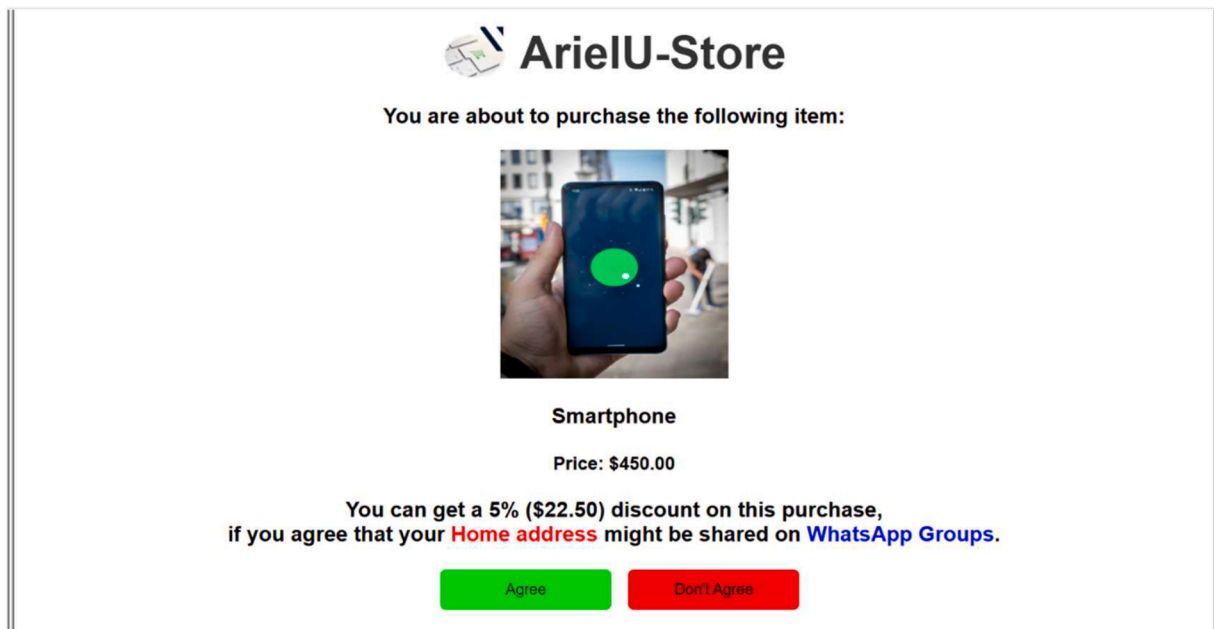


Fig. 2. The advanced stage of negotiation in the last *SI* value of smartphone.

sensitivity to privacy in absolute monetary terms and their willingness to trade privacy relative to the value of the incentive offered.

For each experimental setting, the analysis examined both descriptive and comparative aspects of disclosure behavior. First, the average accepted values were calculated in both percentage and absolute monetary terms to capture relative and concrete privacy valuation. Second, the outcomes were compared across the four main factors to identify which contextual variables most strongly influenced disclosure decisions. Finally, an univariate ANOVA model was applied to test the statistical significance of the main effects and interaction effects among timing, sale item, data item, and platform. This multi-level evaluation enabled combining descriptive interpretation with inferential statistical analysis.

Tables 4 and 5 present the average raw results for the *during* and *after* timing conditions, respectively. Results are shown both in percentages

and absolute terms, across all combinations of the data factors, items and platforms. The totals are provided for reference. The numerical values displayed in the main *Data Item* rows in bold (*Bank account number*, *Email*, *Home address*, *Name*, *Phone number*) represent the averages of all platform-specific results shown in the sub-rows beneath them. These averages summarize the overall valuation for each data item across Facebook, X/Twitter, and WhatsApp - providing a single representative figure per data type.

The results reveal clear differences in how participants valued their personal data across the four experimental factors. Specifically, variations emerged according to the type of data disclosed, the platform of exposure, the product offered in return, and the timing of the discount negotiation. The following subsections present the findings for each factor in detail.

The following sub-sections describe the results of the empirical study

Table 4

The average raw data for the *during* timing results.

Sale Item	Adult toy		Chocolate		Lingerie		Pencil		Smartphone		Total	
	%	\$	%	\$	%	\$	%	\$	%	\$	%	\$
Data Item												
Platform												
Bank account number	16.3	5.7	13.3	2.0	11.7	6.4	16.1	0.3	6.3	28.3	12.7	8.3
Facebook	14.6	5.1	17.5	2.6	18.2	10.0	10.8	0.2	5.0	22.5	13.1	8.5
Twitter	41.7	14.6	17.4	2.6	6.0	3.3	34.0	0.7	4.3	19.5	19.8	6.2
WhatsApp	3.6	1.3	4.4	0.7	4.6	2.5	12.0	0.2	8.6	38.6	7.0	9.5
Email	13.6	4.7	12.1	1.8	15.4	8.5	4.1	0.1	5.9	26.6	10.4	8.7
Facebook	28.4	9.9	8.4	1.3	11.9	6.5	3.3	0.1	5.0	22.5	9.8	9.7
Twitter	9.8	3.4	3.8	0.6	11.6	6.4	4.0	0.1	5.0	22.5	7.8	6.9
WhatsApp	3.5	1.2	25.8	3.9	25.3	13.9	4.7	0.1	11.3	51.0	14.2	9.5
Home address	19.2	6.7	9.4	1.4	18.8	10.4	10.2	0.2	5.8	26.0	12.9	7.8
Facebook	25.5	8.9	14.3	2.2	16.9	9.3	6.3	0.1	7.7	34.5	15.6	9.4
Twitter	13.8	4.8	8.0	1.2	15.8	8.7	13.2	0.3	6.2	27.8	11.1	7.5
WhatsApp	16.7	5.8	3.9	0.6	36.5	20.1	10.9	0.2	3.5	15.8	11.1	5.9
Name	13.5	4.7	14.8	2.2	13.3	7.3	12.7	0.3	5.1	23.0	12.1	6.8
Facebook	3.7	1.3	19.1	2.9	11.6	6.4	19.1	0.4	6.4	28.8	13.6	5.9
Twitter	12.7	4.4	3.5	0.5	22.6	12.4	11.4	0.2	4.8	21.8	12.6	8.1
WhatsApp	20.2	7.1	16.8	2.5	5.6	3.1	5.1	0.1	4.5	20.3	10.2	6.7
Phone number	9.9	3.5	8.6	1.3	13.1	7.2	12.0	0.2	4.9	22.3	9.9	6.9
Facebook	6.5	2.3	19.5	2.9	7.2	4.0	13.2	0.3	6.3	28.1	10.0	6.1
Twitter	16.0	5.6	3.0	0.5	16.9	9.3	17.5	0.4	5.6	25.3	12.5	8.9
WhatsApp	4.7	1.6	6.3	0.9	12.6	6.9	5.5	0.1	3.4	15.4	7.0	5.4
Total	14.9	5.2	11.5	1.7	14.3	7.9	11.2	0.2	5.6	25.2	11.6	7.7

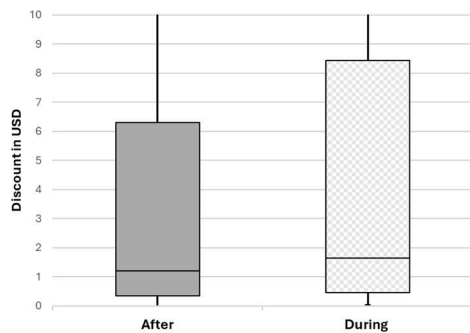
Table 5The average raw data for the *after* timing results.

Sale Item	Adult toy		Chocolate		Lingerie		Pencil		Smartphone		Total	
	%	\$	%	\$	%	\$	%	\$	%	\$	%	\$
Data Item												
Platform												
Bank account number	5.5	15.7	1.4	9.6	7.7	14.0	0.1	5.4	24.6	5.5	8.2	10.2
Facebook	5.5	15.6	0.7	4.4	3.3	6.0	0.2	10.2	23.3	5.2	10.5	7.9
Twitter	5.6	16.1	2.7	18.2	7.8	14.1	0.1	3.3	27.9	6.2	7.7	11.5
WhatsApp	5.3	15.2	0.9	6.2	9.4	17.0	0.0	2.0	22.5	5.0	5.6	11.4
Email	3.0	8.5	1.8	12.3	2.5	4.6	0.2	10.6	22.5	5.0	5.3	8.2
Facebook	2.3	6.5	2.9	19.0	1.9	3.5	1.2	60.0	13.5	3.0	3.9	9.6
Twitter	5.0	14.2	1.4	9.6	2.4	4.3	0.2	9.9	31.5	7.0	7.1	8.9
WhatsApp	2.4	6.8	0.9	5.7	3.4	6.2	0.1	5.1	20.3	4.5	4.6	5.6
Home address	3.0	8.5	0.8	5.6	7.2	13.1	0.2	9.1	23.8	5.3	7.1	8.1
Facebook	1.9	5.3	0.7	4.6	3.9	7.0	0.1	7.1	33.3	7.4	8.0	6.4
Twitter	1.8	5.0	0.6	3.7	13.7	24.8	0.1	4.3	24.3	5.4	9.2	9.8
WhatsApp	3.9	11.2	1.1	7.6	2.9	5.3	0.3	12.5	17.4	3.9	5.2	8.1
Name	3.4	9.8	1.9	12.4	8.4	15.2	0.1	3.9	18.0	4.0	6.7	9.4
Facebook	4.4	12.7	0.6	3.8	12.5	22.7	0.1	4.5	14.4	3.2	6.8	10.2
Twitter	2.6	7.3	0.7	4.5	11.3	20.5	0.1	3.7	22.5	5.0	8.4	8.0
WhatsApp	2.2	6.3	3.3	22.2	3.9	7.0	0.1	3.7	16.9	3.8	5.2	9.8
Phone number	6.0	17.2	1.2	7.9	7.0	12.8	0.1	3.4	36.0	8.0	8.1	10.0
Facebook	8.1	23.0	0.8	5.4	20.1	36.5	0.1	4.0	23.6	5.3	8.2	12.5
Twitter	1.2	3.4	2.1	14.2	6.1	11.0	0.1	2.9	58.5	13.0	9.4	8.1
WhatsApp	10.4	29.6	0.8	5.0	2.5	4.5	0.1	3.7	22.5	5.0	6.0	10.1
Total	4.3	12.4	1.4	9.1	6.5	11.9	0.1	6.7	24.5	5.4	7.1	9.1

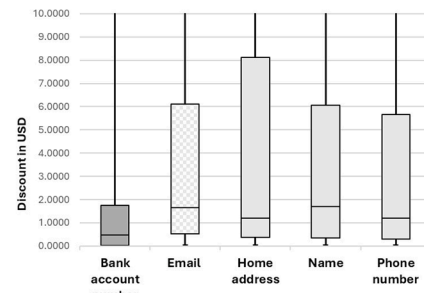
structured around the four experimental factors: (a) Timing of the Discount Offer, (b) Type of Personal Information, (c) Platform of Exposure, and (d) Product Offered. For each factor, we summarize descriptive statistics (tables and boxplots) and interpret the observed trends in the participants' valuations.

4.2.1. Timing of the discount offer

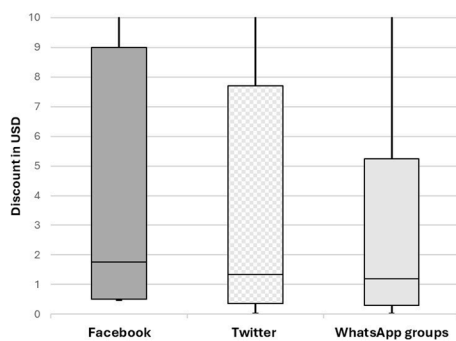
A comparison of the average accepted values before and after users



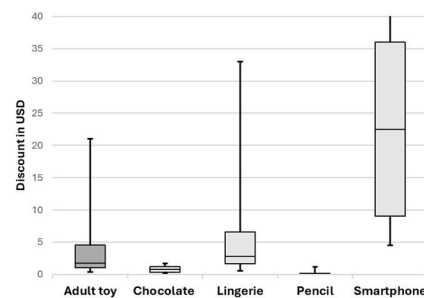
a. Timing of the Discount Offer



b. Personal Data Items



c. Platforms in which the data is exposed



d. Products Offered (Sale Item)

Fig. 3. A stand-alone factor comparison for each of the four different factors of the experimental evaluation.

4.2.2. Type of personal information

Participants demonstrated clear differences in how they valued different types of personal information. As presented in Fig. 3b, bank account numbers received the highest valuations, with a median price of 1.75 indicating a wide range of high-value responses. Surprisingly, names and email addresses also received relatively high valuations, with a median of 1.65, close to that of bank account numbers, suggesting that users perceive email privacy as highly sensitive. Home addresses and phone numbers received similarly high median valuations (1.2), contrary to initial expectations that names would be valued the least. These results suggest that participants may assign non-obvious sensitivity levels to certain data types, possibly influenced by context or framing within the simulation. This distribution challenges conventional assumptions about privacy sensitivity gradients and implies a more nuanced or personalized perception of data value.

4.2.3. Platform of exposure

The perceived sensitivity of data exposure varied significantly by platform. As seen in Fig. 3c, Facebook, with its semi-public sharing model, presented the highest median valuation (1.75), surpassing even the fully public Twitter (1.35). This suggests that participants may perceive Facebook as more intimately tied to their identity, increasing their caution about disclosures. WhatsApp Groups, typically considered the most private environment, yielding the lowest median valuation (1.2). These results indicate that user's factor in not only technical privacy configurations but also the contextual trust they associate with each platform.

These findings imply that platform reputation and context, not just default visibility settings, influence users' valuation of their privacy.

4.2.4. Product offered (Sale item)

The type of product offered in exchange for personal data had a significant effect on the users' willingness to disclose information. As shown in Fig. 3d, the smartphone, representing the most expensive and valuable item, presented the highest average accepted privacy price, with an average valuation of 22.5 and a maximum of 85.5. This indicates a strong association between product value and privacy cost. Users also displayed elevated valuations (average values of 2.75 and 1.75, respectively) in reference to lingerie and adult toys, which may carry social sensitivity, suggesting that emotional or reputational context also influences users' decisions. In contrast, lower-value items such as pencils and chocolates corresponded to a much lower willingness to trade private data. The pencil, for example, had an average accepted value of just 0.1, with a maximum of only 1.2, reinforcing the idea that users are unwilling to risk privacy for minimal incentives. Overall, the results support the hypothesis that users weigh both monetary and social value when deciding to disclose personal information and are more likely to trade privacy when the perceived reward is high.

To assess the statistical significance of the experimental variables influencing users' willingness to disclose personal data, a series of univariate ANOVAs was conducted. The two separate models mentioned were tested: one using the actual monetary valuation in USD (avg-\$) and the other using normalized scale scores (avg), as depicted in Tables 6 and 7, correspondingly. Each model included the four key experimental factors: data item, platform, sale item, and timing (*after/during*), as well as their two-way and higher-order interactions.

Key findings:

- The main effect of the Sale-Item was highly significant ($F = 140.824$, $p < .001$), confirming that the monetary value of the product had the strongest influence on the amount users demanded for their data.
- A significant two-way interaction between After-During and Data-Item ($p = .034$), and a three-way interaction between After-During, Sale-Item, and Platform ($p = .009$), reinforce the conclusion that context and incentive structure jointly shape privacy valuation.

Table 6

All results for the ANOVA on USD-Scaled Values.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
After-During	123.0	1	123.0	1.188	.276
Sale-Item	58,315.5	4	14,578.8	140.824	.000
Data-Item	261.2	4	65.3	0.631	.641
Platform	176.5	2	88.2	0.853	.427
After-During * Sale-Item	49.2	4	12.3	0.119	.976
After-During * Data-Item	1084.6	4	271.1	2.619	.034
After-During * Platform	466.3	2	233.1	2.252	.106
Sale-Item * Data-Item	1315.6	16	82.2	0.794	.693
Sale-Item * Platform	394.1	8	49.2	0.476	.874
Data-Item * Platform	1004.5	8	125.5	1.213	.288

Table 7

All results for the ANOVA on Normalized Scores.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
After-During	1027.5	1	1027.5	3.386	.066
Sale-Item	2012.2	4	503.0	1.658	.158
Data-Item	303.6	4	75.9	0.250	.910
Platform	1086.6	2	543.3	1.790	.168
After-During * Sale-Item	21.6	4	5.4	0.018	.999
After-During * Data-Item	1660.8	4	415.2	1.368	.243
After-During * Platform	78.0	2	39.0	0.129	.879
Sale-Item * Data-Item	4146.5	16	259.1	0.854	.624
Sale-Item * Platform	1622.9	8	202.8	0.668	.720
Data-Item * Platform	4576.3	8	572.0	1.885	.059

- The full four-way interaction was not statistically significant ($p = .221$), although some three-way combinations approached significance (e.g., After-During * Sale-Item * Platform, $p = .009$).

In summary, while user valuation of privacy is highly heterogeneous, the type of product offered (Sale-Item) emerged as the strongest and most consistent predictor of privacy valuation in both normalized and monetary terms. Contextual factors such as timing and platform interacted with product and data type to shape user decisions, particularly in the USD-based model. These findings underscore the importance of aligning incentive structures with privacy risks in designing user-facing aspects of privacy-aware systems.

To strengthen the evaluation, the findings were examined using both raw monetary values and normalized percentage-based values, thereby reducing the risk that the results would have been driven solely by differences in product prices.

4.2.4.1. ANOVA on normalized scores. The second model examined the effects on the normalized privacy valuation (on a scale from 0 to 100). The overall model demonstrated limited explanatory power, with an R^2 of 0.172 and an adjusted R^2 of only 0.013, indicating that much of the variance in user behavior could not be explained by the factors alone.

Key results:

- We did not reveal any standard significance thresholds with reference to the main effects ($p < .05$).
- The only interaction term for which we revealed significance was the three-way interaction between After-During, Sale-Item, and Platform

($p = .030$), suggesting a conditional relationship among timing, product type, and platform contexts.

- We did not reveal any significance with reference to all of the other main and interaction effects ($p > .05$), thus emphasizing the complexity and variability in subjective privacy valuation.

5. Discussion

Prior research on privacy risks in Online Social Networks (OSNs) has primarily focused on identifying technical leakage mechanisms, designing privacy control systems, or proposing theoretical frameworks for privacy valuation. While these studies have contributed significantly to understanding the structural and technical aspects of privacy risks, many lack empirical methods for quantifying how users themselves perceive and value potential privacy losses in practical disclosure scenarios. In particular, several approaches rely on theoretical models, architectural analysis, or survey-based evaluations that do not capture real-time decision behavior when users face concrete trade-offs between privacy and incentives. The approach proposed in this study advances this line of research by introducing a behavioral experimental framework that directly measures users' implicit valuation of personal data through observable disclosure decisions. By simulating realistic incentive-based scenarios, the framework bridges the gap between theoretical privacy valuation models and practical user behavior in OSNs. All of the result files are available on GitHub (Voloch, 2025b).

In this study empirically we estimated the monetary cost of personal data leakage in OSNs, by measuring the compensation users request when facing controlled scenarios of potential information exposure. Through a controlled experimental design using an interactive e-commerce simulation, we were able to evaluate the monetary ranges in which individuals are willing to disclose specific types of personal information in return for perceived benefits. These tangible benefits indicated the perceived value of the data.

5.1. Discussion on results

The empirical experiment included 249 valid participants whose disclosure decisions generated the valuation dataset analyzed in this study. The raw results showed substantial variability across factors, with bank account details receiving the highest average accepted compensation, Facebook exhibiting the highest median valuation among platforms, and high-value products such as smartphones producing the largest increases in users' willingness to accept data leakage or unintended disclosure risks. By embedding negotiation dynamics into the simulation, the users' behavioral thresholds were captured in a way that are not possible via traditional surveys or self-report methods, thereby providing the quantitative backbone of the empirical analysis presented in subsequent sections.

One of the most notable findings of this study is the non-uniformity in privacy valuation. Users did not treat all personal data equally, nor did they respond to exposure risks in a predictable hierarchy of sensitivity. While bank account numbers and home addresses were expected to rank as highly sensitive, and indeed were valued accordingly, it was particularly interesting to observe that more seemingly mundane data, such as names and email addresses, which are not necessarily classified information, were also assigned relatively high valuations. This finding suggests that users may not base their privacy decisions solely on the technical sensitivity of data types, but also on subjective perceptions, prior experiences, or contextual framing of the disclosure request. This pattern is consistent with findings revealed in behavioral privacy research that demonstrate that disclosure decisions are highly context-dependent and shaped by subjective perceptions and framing rather than by any fixed hierarchy of data sensitivity (Acquisti, et al., 2013). This perception also maintains the *privacy paradox*, a well-documented phenomena in the literature, which shows that self-disclosure on social network sites is shaped not only by privacy concerns but also by

individual characteristics and the perceived social relevance of the context (Taddicken, 2014). Another critical dimension uncovered by the experiment is the role of platform context. Although all platforms involved in the study carried some risk of third-party data leakage or unintended disclosure, user responses varied significantly depending on where the disclosure occurred. Facebook for instance, elicited the highest median privacy valuation, even more than the fully public Twitter. This result suggests the psychological or identity-based relationships users have with specific platforms. For example, Boyd and Ellison (2007) define social network sites as services that allow individuals to construct a public or semi-public profile within a bounded system, and indicate that platforms that are more identity-anchored may be guarded more cautiously regardless of their technical privacy settings. Those perceived as more "personal" or identity-based appear to be guarded more cautiously, regardless of their technical privacy settings. Conversely, even platforms assumed to be private, such as WhatsApp groups, did not always lead to lower valuations, indicating that user trust is not solely driven by default access configurations.

A particularly important contribution of the study lies not in showing that incentives influence behavior per se, a fact well established in the behavioral sciences, but in demonstrating how the value and nature of incentives, as quantitatively perceived, may shape users' privacy valuation in data-leakage scenarios within OSNs. While prior work has shown that external incentives affect decision-making by altering perceived costs and benefits (Kruglanski, et al., 2015), such effects have rarely been measured in the context of privacy disclosure, nor operationalized as an explicit monetary cost of personal data. Our findings show that incentive magnitude and social sensitivity systematically modulate the price users demand for potential data exposure, thereby linking abstract motivational theory to measurable privacy tradeoffs in realistic OSN settings. As hypothesized, the value of the product offered in exchange for personal data was a strong predictor of the willingness to disclose. High-value items such as smartphones resulted in significantly higher levels of acceptance, while low-value items like pencils or chocolates corresponded with minimal data disclosure. Moreover, products associated with personal or social sensitivity, such as lingerie, also led to elevated valuations, suggesting that reputational risk plays a measurable role in shaping privacy decisions.

5.2. Practical implications

Beyond the empirical findings, the results of this study have several implications for the design of digital platforms and privacy-aware systems. First, the results demonstrate that users do not assign a fixed value to their personal data; rather, privacy valuation is highly context-dependent and influenced by incentive structures and platform environments. This suggests that static privacy controls or one-size-fits-all consent mechanisms may be insufficient to capture the diversity of user preferences, an insight that is crucial to design privacy protection mechanisms. Second, the findings highlight how economic incentives can influence disclosure behavior, even when users recognize potential privacy risks. This dynamic may contribute to environments where platforms indirectly encourage data sharing through reward structures, discounts, or convenience features. Understanding these mechanisms is important for designing more transparent consent processes and for evaluating whether incentive-based disclosure practices align with user interests. Third, the results suggest that privacy-aware system design may benefit from adaptive mechanisms that consider contextual factors when prompting users about information disclosure. For example, privacy decision-support tools or AI-based assistants could help users better evaluate the potential trade-offs between short-term rewards and long-term privacy implications.

From a policy perspective, these findings reinforce the importance of transparency and user awareness in digital ecosystems where personal data is exchanged for services or incentives. By quantifying how users value their personal data under different contexts, this study contributes

empirical evidence that may support the development of privacy regulations, governance strategies, and privacy protection tools.

5.3. Limitations

While the methodology, the experimental design and the analysis provide valuable insights, some limitations should be acknowledged. First, the simulation was based on hypothetical data disclosure in a controlled environment, which may not fully capture the emotional, social, and habitual factors in real-world OSN behavior. Users may behave differently when actual social visibility or peer judgment is involved. Second, the study did not take into account individual differences in privacy literacy, risk aversion, or digital habits, which could explain some of the behavioral heterogeneity observed.

For example, previous research showed that age-related differences in social media use, privacy literacy and privacy attitudes systematically shape both disclosure and protective behaviors on Facebook (Kezer, et al., 2016). Other works indicate that online privacy literacy shapes users' ability and willingness to engage in protective behaviors (Trepte, et al., 2014) and that individual risk attitudes and perceptions of privacy risk systematically influence disclosure decisions (Hermes, et al., 2020). Incorporating such individual-level factors into future models could help explain part of the results' diversity in the current study. Third, while the product and platform choices aim to reflect realistic scenarios, their interpretations by users may have varied depending on demographic or cultural background. Finally, the short-term nature of the experiment does not address the long-term consequences of data disclosure or cumulative risk over time, which is a key consideration in understanding the full impact of privacy related decisions.

Despite these limitations, the study provides several scientific and practical contributions. It proposes a solid framework to estimate privacy valuations in real-time decision contexts, offering a model that can be applied by designers, regulators, and OSN operators to align user incentives with privacy-respecting practices. For example, platforms might implement adaptive privacy prompts that are scaled based on the perceived sensitivity of both the context and the incentive or introduce dynamic controls that account for the emotional and behavioral drives concerning privacy trade-offs. This direction aligns with prior evidence that interface-level 'privacy nudges' (Wang, et al., 2014) can steer users toward more deliberate, context-sensitive disclosure decisions in OSN settings. In other words, the methodology that was developed and evaluated in this research, may be a central component of an AI-agent that will assist users in making adequate information disclosing decisions. The research results suggest that one-size-fits-all privacy settings are insufficient to capture the diversity of user preferences, and that more nuanced, customizable solutions are needed. This insight may have a significant impact on the design of new systems, and the creation of privacy regulations.

Another limitation relates to information asymmetry between users and platforms, a phenomenon widely discussed in privacy economics (Acquisti, et al., 2013). In real OSN environments, users typically do not have full visibility into how their data may be aggregated, analyzed, or monetized by platforms and third parties. Consequently, the valuations observed in the experiment may reflect users' perceived immediate costs of disclosure rather than the full long-term economic value of their personal data.

5.4. Future work

Future work could expand this model by integrating longitudinal data, exploring cross-cultural differences, or incorporating biometric and behavioral analytics to refine sensitivity models. Longitudinal effects are especially relevant in OSNs, where privacy preferences and acceptable information flows can change as content ages (Ayalon and Toch, 2013). Additional studies might also simulate social pressure, peer effects, or public visibility to better approximate the OSN environment.

This however, require an altering of the core model beyond the curation of new datasets.

Several more avenues for future research emerge from this study. The experimental design could expand by including additional social platforms, broader demographic groups, or alternative incentive structures to explore how cultural, social, or economic factors influence privacy valuation. Further studies could investigate long-term privacy perceptions by examining how repeated disclosure decisions evolve over time as users become more aware of potential risks. Another aspect is integrating machine learning or adaptive experimental designs could enable dynamic estimation of privacy valuation across larger and more diverse populations.

Ultimately, understanding how users perceive the cost of their personal data, and how that perception shifts across contexts, is crucial for building ethical, transparent, and user-aligned digital ecosystems. The results shed light on several important dynamics underlying user behavior in digital environments, with direct implications for privacy policy design, user interface engineering, and OSN governance.

5.5. Conclusion

This study investigated how users of OSNs assign economic value to their personal information when confronted with disclosure incentives. While previous research has primarily focused on technical privacy risks, leakage mechanisms, or privacy attitudes, fewer studies have examined how users implicitly value their personal data in realistic disclosure scenarios. To address this gap, a behavioral experimental methodological framework is proposed that estimates privacy valuation from observable user decisions. The methodology was implemented through an interactive simulation in which participants received discount offers in exchange for disclosing various types of personal information. The experimental design incorporated four contextual factors: the type of personal data requested, the exposure platform, the product offered as an incentive, and the timing of the discount offer. Using responses collected from 249 participants, the study estimated users' privacy valuations by analyzing the compensation thresholds at which individuals were willing to disclose personal data.

The results demonstrate that users assign significantly different values to different types of personal information. Financial information, such as bank account details, was associated with the highest privacy valuation. At the same time, contextual elements such as the platform of exposure and the nature of the incentive also influenced disclosure decisions. In particular, higher-value or socially sensitive products increased users' willingness to disclose personal data, highlighting the contextual and behavioral nature of privacy decisions in OSN environments. Beyond the empirical findings, this work contributes a methodological framework for studying privacy valuation through behavioral experimentation. By modeling privacy disclosure as a negotiation process between incentives and perceived privacy cost, the proposed approach provides a practical mechanism for estimating how individuals evaluate privacy trade-offs in digital environments.

The findings of this study contribute to the literature on privacy and data valuation in several important ways. First, the study introduces a behavioral experimental approach for estimating the monetary value users assign to personal information in OSNs. Unlike survey-based approaches that rely on self-reported attitudes, the proposed methodology derives privacy valuations from observable decision behavior in controlled disclosure scenarios. Second, the research demonstrates that privacy valuation is highly contextual. The results show that disclosure decisions are influenced not only by the sensitivity of the data itself but also by contextual factors such as the platform of exposure, the type of incentive offered, and the timing of the offer. This finding highlights the importance of considering environmental and situational factors when designing privacy policies and consent mechanisms. Third, the experimental framework provides a practical methodology for studying privacy decision-making under realistic incentive structures. By integrating

a simulated environment with iterative negotiation mechanisms, the study enables a more realistic representation of how users evaluate privacy trade-offs in digital ecosystems. Finally, the study provides empirical evidence to inform the design of privacy-aware systems, data governance policies, and user-centric privacy tools. Understanding how users value their personal data economically can help platform designers and regulators better align privacy protections with users' actual perceptions and behaviors. This insight is especially crucial for the development of an AI-based agent aimed at supporting user decisions concerning publishing data in the complex digital world.

CRedit authorship contribution statement

Nadav Voloch: Writing – original draft, Validation, Software, Methodology, Investigation. **Ron S. Hirschprung:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Acquisti, A., John, L.K., Loewenstein, G., 2013. What is privacy worth? *J. Leg. Stud.* 42 (2), 249–274.
- Acquisti, A., Taylor, C., Wagman, L., 2016. The economics of privacy. *J. Econ. Lit.* 54 (2), 442–492.
- Ayalon, O., Toch, E., 2013. Retrospective privacy: managing longitudinal privacy in online social networks. In: *Proceedings of the Ninth Symposium on Usable Privacy and Security*, pp. 1–13.
- Boyd, D.M., Ellison, N.B., 2007a. Social network sites: definition, history, and scholarship. *J. Comput.-Mediat. Commun.* 13 (1), 210–230.
- Boyd, D.M., Ellison, N.B., 2007b. Social network sites: definition, history, and scholarship. *J. Comput.-Mediat. Commun.* 13 (1), 210–230.
- Cerruto, F., et al., 2022. Social network data analysis to highlight privacy threats in sharing data. *J. Big. Data* 9 (1), 19.
- Cirillo, S., et al., 2023. A Visual Privacy Tool to Help Users in Preserving Social Network Data. *IS-EUD Workshops* s.l.
- D'Annunzio, A., Menichelli, E., 2022. A market for digital privacy: consumers' willingness to trade personal data and money. *J. Ind. Bus. Econ.* 49 (3), 571–598.
- Dibia, O. et al., 2024. SoK: usability studies in Differential Privac. *arXiv preprint arXiv:2412.16825*.
- Galperti, S., Perego, J., 2023. Privacy and the value of data. In: *AEA Papers and Proceedings*, 113, pp. 197–203.
- Hermes, S., et al., 2020. Objective versus relative risk in privacy decision making: a replication study from Germany. *AIS Trans. Replication Res.* 6 (1), 11.
- Hirschprung, R., Toch, E., Bolton, F., Maimon, O., 2016. A methodology for estimating the value of privacy in information disclosure systems. *Comput Hum. Behav.* 61, 443–453.
- John, L.K., Acquisti, A., Loewenstein, G., 2011. Strangers on a plane: context-dependent willingness to divulge sensitive information. *J. Consum. Res.* 37 (5), 858–873.
- Kezer, M., Sevi, B., Cemalcilar, Z., Baruh, L., 2016. Age differences in privacy attitudes, literacy and privacy management on Facebook. *Cyberpsychology: J. Psychosoc. Res. Cybersp.* 10 (1).
- Kim, J., Kim, J., Chang, H., 2020. Research on behavior-based data leakage incidents for the sustainable growth of an organization. *Sustainability.* 12 (15), 6217.
- Krishnamurthy, B. & Wills, C.E., 2009. *On the leakage of personally identifiable information via online social networks*. s.l., s.n., pp. 7–12.
- Kross, E. et al., 2013. Facebook use predicts declines in subjective well-being in young adults. *8*(8), p. e69841.
- Kruglanski, A.W., et al., 2015. The architecture of goal systems: multifinality, equifinality, and counterfinality in means-end relations. *Adv. Motiv. Sci.* 69–98.
- Lindamood, J., Heatherly, R., Kantarcioglu, M. & Thuraisingham, B., 2009. *Inferring private information using social network data*. s.l., s.n., pp. 1145–1146.
- Li, Y., Li, Y., Yan, Q., Deng, R.H., 2015. Privacy leakage analysis in online social networks. *Comput. Secur.* 239–254.
- McDonald, N., Forte, A., 2021. Powerful privacy norms in social network discourse. In: *Proceedings of the ACM on Human-Computer Interaction*, 5, pp. 1–27.
- Mitchell, D., El-Gayar, O.F., 2022. Privacy and Online Social Networks: a systematic literature review of concerns. *Preserv. Polic.*
- Molok, N.N.A., Ahmad, A., Chang, S., 2018. A case analysis of securing organisations against information leakage through online social networking. *Int. J. Inf. Manage* 43, 351–356.
- Nayak, S.K., Ojha, A.C., 2019. *Data Leakage Detection and Prevention: Review and Research Directions*. Springer, pp. 203–212 s.l.
- Praveena, A., Smyth, S., 2016. Anonymization in Social Networks: a survey on the issues of data privacy. *J. Int. J. Eng. Comput. Sci.* 5 (3), 15912–15918.
- Prince, C., Omrani, N., Schiavone, F., 2024. Online privacy literacy and users' information privacy empowerment: the case of GDPR in Europe. *Inf. Technol. People* 37 (8), 1–24.
- Rivera, M., 2025. 28 Social Media statistics marketers and creators need to know for 2025 [Online] Available at: <https://www.clearvoice.com/resources/social-media-statistics>.
- Skatova, A., McDonald, R., Ma, S., Maple, C., 2023. Unpacking privacy: valuation of personal data protection. *PLoS. One* 18 (5), e0284581.
- Sood, K. et al., 2025. A framework to prevent biometric data leakage in the Immersive technologies domain. *arXiv preprint arXiv:2505.04123*.
- Srivastava, A., Geethakumari, G., 2013. Measuring Privacy Leaks in Online Social Networks. *IEEE*, pp. 2095–2100 s.l.
- Sun, Y., Ma, X., Han, A., 2025. MAB-RSP: data pricing based on Stackelberg game in MCS. *Array* 26, 100380.
- Taddicken, M., 2014. The "privacy paradox" in the Social web: the impact of privacy concerns, individual characteristics, and the perceived social relevance on different forms of self-disclosure. *J. Comput.-Mediat. Commun.* 19 (2), 248–273.
- Tomy, S., Pardede, E., 2016. Controlling privacy disclosure of third party applications in online social networks. *Int. J. Web Inf. Syst.* 12 (2), 215–241.
- Trepte, S., et al., 2014. Do people know about privacy and data protection strategies? Towards the "Online Privacy Literacy Scale"(OPLIS). *Reforming Eur. data prot. law* 333–365.
- Voloch, N., 2025a. *DataLeakageOSNResults* [Online] Available at: <https://github.com/nadavvoloch/DataLeakageOSNResults>.
- Voloch, N., 2025b. *OSN data leakage privacy simulation on GitHub* [Online] Available at: <https://github.com/nadavvoloch/DataLeakageOSN>.
- Voloch, N., Gal-Oz, N., Gudes, E., 2021. A trust based privacy providing model for online social networks. *Online Soc. Netw. Media* 24, 100138.
- Wang, Y., et al., 2014. A field trial of privacy nudges for facebook. In: *Proceedings of the SIGCHI conference on human factors in computing systems*, pp. 2367–2376.
- Watson, L., Andreeva, R., Yang, H.-T. & Sarkar, R., 2022. Differentially private Shapley values for data evaluation. *arXiv preprint arXiv:2206.00511*.
- Wong, W.P., Tan, H.C., Tan, K.H., Tseng, M.-L., 2019. Human factors in information leakage: mitigation strategies for information sharing integrity. *Ind. Manag. Data Syst.* 119 (6), 1242–1267.
- Zhou, Y., Piramuthu, S., 2015. Information re-sharing in Social Media: a conceptual framework of third-party Information leakage. *Decis. Support. Syst.* 73, 77–89.